Sanjana R

rsanjana2027@email.com | LinkedIn | GitHub | LeetCode | +91 90359 83960 | Bengaluru, India |

Summary

AIML undergraduate with strong interest in Artificial Intelligence and Machine Learning. Hands-on experience in Python, SQL, and developing AI/ML-based applications and software projects. Passionate about problem-solving, modern AI technologies, and building scalable intelligent solutions.

Experience

AIML Intern – Emberquest Jan 2024 – May 2024

- Completed a 16-week internship focused on Artificial Intelligence and Machine Learning concepts.
- Worked on Python-based AI/ML tasks involving data preprocessing, model implementation, and basic predictive analysis using real-world datasets.
- Gained hands-on experience in Artificial Intelligence and Machine Learning through 16-week practical assignments and project-based learning activities.

Education

BMS Institute of Technology and Management Sep 2024 – June 2027
Bachelor of Engineering – Artificial Intelligence and Machine Learning

M N Technical Institute Oct 2021 – May 2024
Diploma in Computer Science
CGPA: 9.23

Skills

- **Languages & Tools:** Python, Java, SQL, Git, GitHub
- **AI & ML:** Machine Learning, Deep Learning, Federated Learning, Computer Vision
- **Frameworks & Libraries:** PyTorch, TensorFlow, NumPy, Pandas, Streamlit, Scikit-learn, OpenCV, MediaPipe

Projects

• FedGDA – Federated Medical Image Segmentation - GitHub

Technologies: Python | PyTorch | NumPy | Deep Learning | Federated Learning
• Developed a federated learning pipeline for multi-organ CT image segmentation using Attention U-Net, achieving Dice Scores of 0.9069 (centralized) and 0.8467 (FedGDA).
• Evaluated a federated learning framework using FedAvg and non-IID client datasets for multi-organ CT image segmentation, client-wise performance variation, and segmentation consistency across heterogeneous data distributions.

• SAMVAAD – Multimodal Indian Sign Language Translator - GitHub

Technologies: Python | TensorFlow | MediaPipe | OpenCV | Streamlit | SQLite
• Developed a multimodal Indian Sign Language translation system enabling sign-to-text, speech-to-sign, and text-to-sign conversion using TensorFlow and MediaPipe with real-time gesture interpretation.
• Implemented hand gesture recognition and landmark tracking using OpenCV and MediaPipe, supporting recognition of multiple ISL gestures with confidence-based prediction and real-time translation.

• Fraud Detection System using Machine Learning - GitHub

Technologies: Python | Scikit-learn | Pandas | NumPy | Matplotlib | SMOTE
• Developed a credit card fraud detection system using Random Forest and SMOTE to identify fraudulent transactions in highly imbalanced financial datasets.
• Built an interactive Streamlit dashboard with fraud probability prediction, risk analysis, confusion matrix, ROC-AUC, precision, recall, F1-score, and feature importance visualization.

Certifications

- **NPTEL** – Business Fundamentals for Entrepreneurs | • **NPTEL** – Data Analytics Using Python
- **Snowflake** – SnowPro Associate: Platform | • **Cisco Networking Academy** – Networking Basics